

Human Capital, Competition, and Mobility in the Managerial Labor Market

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- ▶ Is CEO pay set by competition among employers, or rents conceded by boards?
 - ▶ *Competition* (Gabaix & Landier 2008): boards pay what a rival would; high pay is a market outcome
 - ▶ *Power* (Bebchuk & Fried 2004): managers take what captured boards tolerate; high pay is a governance failure
- ▶ The market for CEOs is strikingly rigid (Cziraki & Jenter 2024)
 - ▶ > 70% promoted internally; < 1% lateral moves per year (S&P 1500)
- ▶ But rigidity is consistent with two very different worlds:
 - ▶ *Competition absent*: offers rarely arrive or aren't worth taking; pay is left to the negotiating table
 - ▶ *Competition invisible*: offers do arrive, but incumbents counter (match, benchmark), so competition ends in raises, not moves
- ▶ Same mobility, opposite verdicts. The difference is visible only in pay

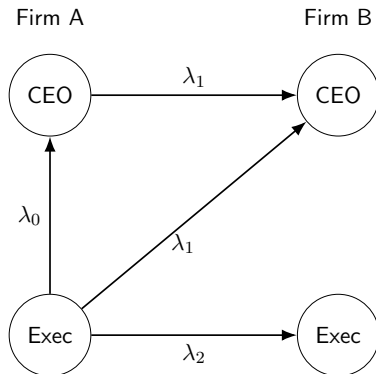
- ▶ We pose and estimate a model of a frictional executive labor market
 - ▶ Competing job opportunities arrive occasionally
 - ▶ Managers accumulate general and firm-specific human capital
 - ▶ Pay is a piece-rate contract, set by bilateral bargaining and bidding among rival employers
- ▶ The model separates pay, in levels and in growth, into its three sources:
 - ▶ **Productivity**: what the manager produces from ability, accumulated human capital, firm technology
 - ▶ **Power**: rents extracted by the manager in bilateral bargaining
 - ▶ **Competition**: the manager's share of output, bid up by rival employers (benchmarking, retention, matched offers)
- ▶ Estimated on a manager-firm panel, 1992-2023 (S&P 1500)

- ▶ Competition is invisible, not absent: explains $\approx 42\%$ of managerial rents
- ▶ Managerial bargaining power is modest ($\hat{\beta} \approx .3$), in line with rank-and-file workers
 - ▶ Notably lower than in previous papers (ex. Taylor 2013, $\approx .5$)
- ▶ Realized rents are 54.6% of match surplus, against 31.6% in a no-competition benchmark
- ▶ Competitive discipline is real, though poaching is rarely realized in equilibrium
 - ▶ Why? Human capital is largely **firm-specific**: a long-tenured manager forfeits $\approx 53\%$ of accumulated human capital by moving
 - ▶ Firms retain firm-specific talent via aggressive competitive benchmarking

- ▶ **The market for CEOs.** Gabaix and Landier (2008), Terviö (2008), Edmans, Gabaix, and Landier (2009); Murphy and Zabojnik (2007), Custódio, Ferreira, and Matos (2013); Frydman and Jenter (2010), Cziraki and Jenter (2024), He and Schroth (2024), Geelen and Hajda (2024), Gompers et al. (2023), Sauvagnat and Schivardi (2024)
- ▶ **Managerial power and rents.** Bebchuk and Fried (2004), Graham, Li, and Qiu (2012), Taylor (2013)
- ▶ **Structural estimation of executive careers.** Taylor (2010), Page (2018), Gao, Wang, and Wu (2021), Ma, Wang, and Wu (2023)
- ▶ **On-the-job search and bargaining.** Becker (1962), Lazear (2009); Postel-Vinay and Robin (2002), Cahuc, Postel-Vinay, and Robin (2006), Bagger et al. (2014)

Model

- ▶ Frictional labor market consisting of firms and managers
 - ▶ Time is continuous
- ▶ Managers find themselves in one of three employment states:
 1. Employed as CEO (C)
 2. Employed as non-CEO executive (E)
 3. Unattached (U)
- ▶ Managers occasionally receive competing job offers
 - ▶ May be an internal promotion or a position at another firm
 - ▶ All labor market shocks follow Poisson processes



- ▶ Job offers trigger a bargaining game, outcome of which determines who gets the manager and at what price
- ▶ Additionally: retirement, movement in/out of unattachment

- ▶ Given labor market experience t and tenure $\tau \leq t$, (log) output per manager is:

$$y = \overbrace{a + g(t)}^{\text{Portable}} + \underbrace{f + k(\tau)}_{\text{Not portable}} \quad (1)$$

- ▶ a - Inherent managerial ability $\sim N(0, \sigma_a^2)$
- ▶ $g(t)$ - General component of human capital
- ▶ f - firm-specific heterogeneity (size, productivity, culture, etc.)
- ▶ $k(\tau)$ - Firm-specific component of human capital
 - ▶ Resets to $k(0) = 0$ any time a manager switches firms

- ▶ Managers and firms bargain over piece rate contracts which specify the share $R \in (0, 1]$ of output awarded to the manager:

$$W = R \times Y \quad (2)$$

- ▶ It is convenient to talk in terms of logs:

$$\begin{aligned} w &= r + y \\ &= r + a + g(t) + f + k(\tau) \end{aligned} \quad (3)$$

- ▶ Note that $r = \log(R) \leq 0$
 - ▶ $r = 0 \iff R = 1$, compensation = marginal product
 - ▶ Perfect competition benchmark

- ▶ An internal CEO offer is negotiated bilaterally between manager and firm
- ▶ New piece rate r' splits promotion surplus via bargaining power $\beta \in [0, 1]$:

$$V_C(r', \tau, f) = V_E(r, \tau, f) + \beta(V_C(0, \tau, f) - V_E(0, \tau, f)) \quad (4)$$

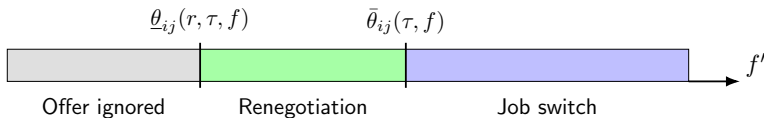
- ▶ Tenure and firm-specific capital $k(\tau)$ continue uninterrupted, f remains fixed
- ▶ External offers differ in two ways:
 - ▶ The poacher's productivity is a fresh draw $f' \sim \Psi(\cdot | a)$
 - ▶ Better managers meet more productive firms (assortative matching)
 - ▶ The incumbent can counter: bargaining becomes a bidding war (next slides)

Compensation and Bargaining

Outside Offers and Their Outcomes

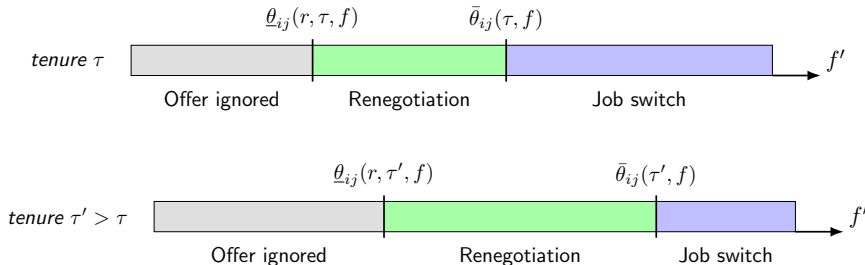
WARWICK

- ▶ An outside offer from a firm of productivity f' triggers a bargaining game between manager, incumbent, and poacher
 - ▶ Bertrand competition between incumbent and poacher
 - ▶ Loser bids full surplus in equilibrium, manager captures a share β of surplus beyond it
- ▶ Three possible outcomes, depending on where f' falls:



Compensation and Bargaining

Firm-Specific Capital Shifts the Regions



- ▶ **Job lock:** as k accumulates, $\bar{\theta}_{ij}$ rises and switching becomes rare
- ▶ Renegotiation region widens; incumbents match a growing mass of offers
 - ▶ Job switches are observable, renegotiation is not
 - ▶ “Invisible” influence of competition grows more important with tenure

Estimation & Identification

Identification

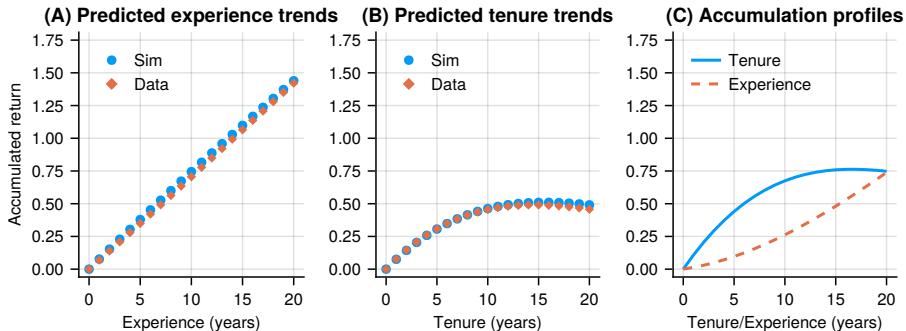
The Piece Rate is Unobserved, and Endogenous

- Pay in the model:

$$w_{it} = a + g(t) + k(\tau) + f + \underbrace{r_{it}}_{\text{contract}} + \epsilon_{it}$$

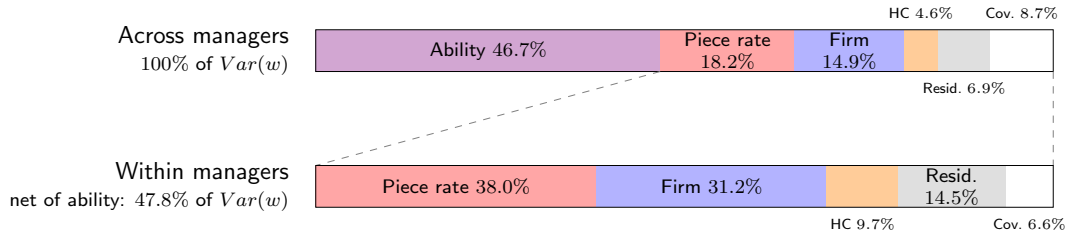
- Four terms are standard Mincer-AKM ingredients. The fifth, r_{it} , is a sufficient statistic for the manager's competitive history
 - Ratchets up at each matched offer, resets at each switch
 - Two identical managers at the same firm can be paid differently: their contracts remember different histories of competition
- But r_{it} is unobserved and endogenous; TWFE regression omitting r_{it} is misspecified and biased
 - SMM: choose parameters so that the same regression, estimated on model-simulated data, replicates this bias

- ▶ **Arrival rates:** duration of job spells, by transition type
- ▶ $g(t)$ **vs** $k(\tau)$: pay profiles in experience vs tenure
 - ▶ Movers reset tenure but not experience: switchers separate the two clocks
- ▶ **Bargaining power** β : size of pay changes at observed transitions
 - ▶ The wedge between pay-level and pay-growth profiles is highly sensitive to β
- ▶ **Ability, firm, and sorting:** moments of manager and firm fixed effects
 - ▶ Slope of firm effects on firm size anchors productivity; slope of *manager* effects on firm size reveals sorting
- ▶ 43 moments, 22 parameters. Matched manager-firm panel spanning 1992-2023 (ExecuComp)



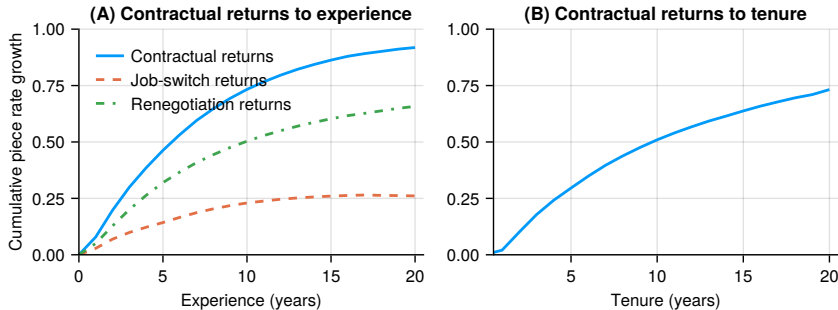
- Structural accumulation profiles differ from reduced-form trends
 - The gap is the estimated bias from the omitted contractual term
 - Switching resets tenure while competition raises pay, tilting the experience profile up and tenure profile down

Results



- ▶ Ability (manager fixed effects) dominate levels
 - ▶ Contrary to frictionless assignment models (ex. Gabaix & Landier 2008)
 - ▶ In line with reduced-form evidence (ex. Graham, Li, & Qiu 2012)
- ▶ Piece rate (contractual history) dominates dynamics
 - ▶ Accumulated record of offers fielded, switches made, route into the CEO seat

Contractual Growth: Renegotiation vs Switching



- ▶ Experience profile: renegotiation delivers 72.8% of career contractual gains
- ▶ Tenure profile (within firm): renegotiation delivers 0.75 log points of pay growth by tenure 20
 - ▶ On par with firm-specific skill accumulation itself

- ▶ Estimate of managerial bargaining power is ordinary: $\hat{\beta} = \mathbf{0.32}$ (0.04)
 - ▶ Comparable to rank-and-file workers (Bagger et al. 2015)
- ▶ Yet, realized surplus capture is much larger: 54.6%
- ▶ The gap is competition. Shutting off bidding, implied bargaining power would be:

$$b = 1 + \frac{r}{f + k(\tau) - f_{min}} \in [\beta, 1] \quad (5)$$

- ▶ Comparing β against b allows us to decompose captured surplus into:
 - ▶ Competition component - $\frac{b-\beta}{b}$
 - ▶ Pure bargaining power component - $\frac{\beta}{b}$

	Realized capture b	Share from competition
Full sample	54.3%	42.7%
Internal CEOs	49.6%	36.2%
Poached CEOs	79.3%	60.1%

- ▶ Ignoring labor market competition, b would be our estimate of bargaining power
 - ▶ Biased upwards for all CEOs ($b > \beta = 0.32$)
- ▶ Poached CEOs capture the most surplus, and benefit the most from competition
 - ▶ Competitive bidding significantly increases the cost of external hires
- ▶ Internal appointees are more insulated from competition

Route to the CEO seat	Baseline	Portable k	Δ
Internal promotion	68.1%	63.7%	-4.4 p.p.
Poached executive	24.8%	27.1%	+2.3 p.p.
Poached CEO	7.2%	9.2%	+2.0 p.p.

- ▶ Counterfactual: make firm-specific capital *fully portable*
- ▶ Absent FSHC, internal dominance is sustained:
 - ▶ Poaching is contested; promotion is not
 - ▶ Internal promotions constitute a **path of least resistance** (internal hires capture 49.6% of surplus, poached CEOs 79.3%)

Conclusion

- ▶ CEOs are paid above their reservation value. How much of this excess stems from competitive discipline?
 - ▶ 42% overall: 36.2% for internal hires, 60.1% for external hires
- ▶ Ignoring labor market condition would misattribute these shares, inflating estimates of managerial power and agency frictions inferred from pay
- ▶ Firm-specific capital makes offers costly to accept, so incumbents retain by matching: competition arrives as raises, not moves (72.8% of contractual gains)
- ▶ Lack of mobility & dominance of internal appointees are the equilibrium consequence of competition, not evidence of its absence